Bachelor Thesis in Algorithmic Game Theory - Lukas Chudy

I'm building on [Calvano et al. (2020)], who showed that AI pricing algorithms can learn to collude autonomously, essentially, when algorithms have memory and patience, they discover high-price strategies without ever communicating. The baseline case (no memory, no patience) yields competitive pricing.

Their model relies on idealized assumptions: synchronous price setting, perfect information, and unlimited memory. My contribution is to test whether collusion survives under realistic frictions, specifically information latency (delayed price observations) and bounded memory (limited recall of past periods). I'll translate their Fortran code to C++, add these constraints, and ideally identify the threshold where collusion breaks down. Even without a sharp threshold, the results will improve our understanding of when algorithmic collusion is sustainable and the policy implications are towards antitrust rules.

Current progress is that I've run the Fortran code on a smaller test set to establish a baseline for verifying my C++ translation.

Timeline:

Now – Mar 7: Translation (Fortran → C++)
Mar 7 – Mar 17: Verification (match Fortran output)
Mar 17 – Apr 5: Extensions (add latency + memory parameters)
Apr 5 – Apr 28: Experiments (run parameter sweeps)
Apr 28 – May 12: Analysis (figures, results interpretation)
May 12 – Jun 12: Writing (full thesis, incorporate feedback)

If I hit roadblocks in the first 2–3 weeks, I can pivot to extending the Fortran code directly rather than doing a full conversion.